

Traffic Sign Classification

Deep Feature Manifolds & Hybrid SVM Optimization

Đỗ Duy Toàn, Vũ Tùng Lâm, Nguyễn Thái An,
Nguyễn Văn Linh, Phạm Việt Hải Đăng, Tạ Hiếu Nam

University of Science and Technology of Hanoi

Final Project - 2026



Outline

- 1 Introduction
- 2 Methodology
- 3 Manifold Strategy
- 4 Results
- 5 Conclusion

Project Objectives

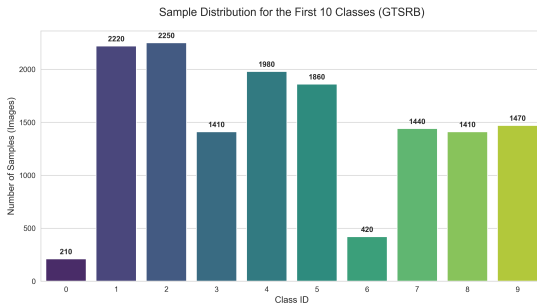
- **Goal:** Build a robust traffic sign recognition system for autonomous vehicles.
- **Key Approach:** Bridge the gap between Deep Learning and Support Vector Machines.
- **Innovation:** Investigation of Low-Dimensional (3D) Manifold Classification.

Dataset: GTSRB (10-class Subset)



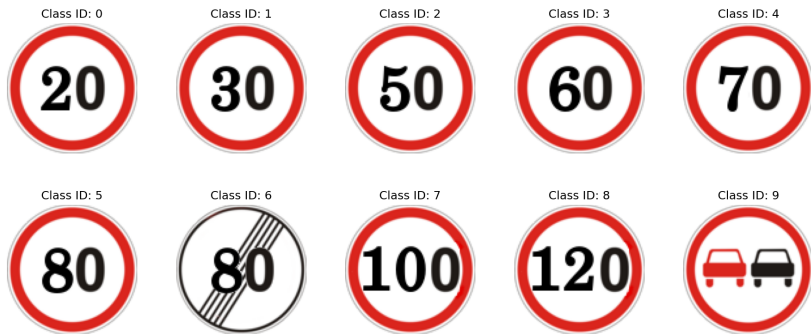
- Over 10,000 images across 10 distinct traffic sign categories.
- Significant variations in illumination, scale, and environmental noise.

Class Distribution



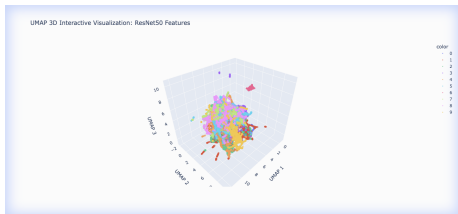
- Observable class imbalance (e.g., speed limits vs. unique signs).
- Addressed using **balanced class weights** in SVM.

Pipeline Architecture

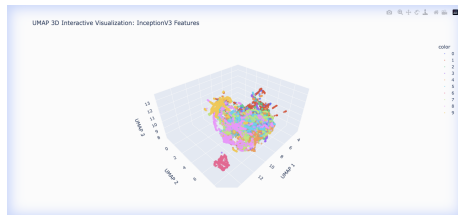


- 1 **Feature Extraction:** ResNet50 & InceptionV3 backbones.
- 2 **Dimensionality Reduction:** PCA & UMAP (3D).
- 3 **Classification:** Hybrid SVM (Linear vs. RBF kernels).

3D Manifold Visualizations (UMAP)



ResNet50 Manifold



InceptionV3 Manifold

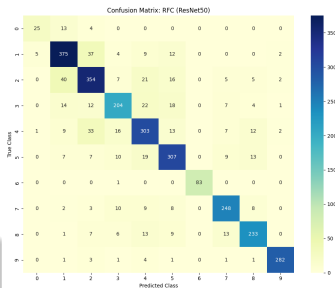
UMAP preserves local neighborhood structures, forming dense clusters.

Classification in 3D Space

- **Discovery:** Linear models (SGDC) fail in 3D ($<35\%$).
- **Solution:** **Random Forest** dominates.
- **Why?** Recursive partitioning isolates non-linear clusters efficiently.

Result (RFC 3D)

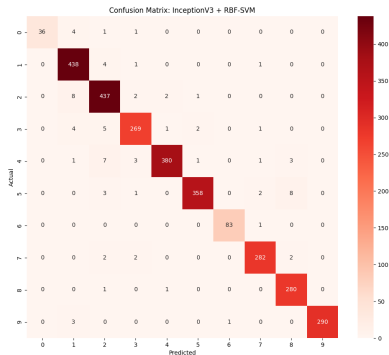
82.28% Accuracy (ResNet50).



RFC Confusion Matrix

High-D Classification Results

Model Configuration	ResNet50	InceptionV3
SGD-SVM (Linear)	91.24%	88.50%
RBF-SVM (Hybrid)	95.91%	97.24%



Peak Model CM (InceptionV3 +
RBF)

Key Takeaways

- **Feature Power:** Pre-trained CNNs provide robust 2048-D manifolds.
- **SVM Hybrid:** RBF kernels are essential for handling non-linear deep features.
- **Manifold Density:** 3D projections retain significant discriminative info.

Project Repository:

<https://github.com/toando0212/lab-ml2>

Thank You!

Q&A